

(a)

| Structure Type                 | Material           | Materials Project ID | Space Group  |
|--------------------------------|--------------------|----------------------|--------------|
| Rock salt (MO)                 | NiO                | mp-19009             | $Fm\bar{3}m$ |
|                                | CoO                | mp-19128             | $Fm\bar{3}m$ |
|                                | FeO                | mp-18905             | $Fm\bar{3}m$ |
| Rutile (MO <sub>2</sub> )      | TiO <sub>2</sub>   | mp-2657              | $P4_2/mnm$   |
|                                | RuO <sub>2</sub>   | mp-2164              | $P4_2/mnm$   |
|                                | IrO <sub>2</sub>   | Mp-2723              | $P4_2/mnm$   |
| Perovskite (ABO <sub>3</sub> ) | SrTiO <sub>3</sub> | mp-5229              | $Pm\bar{3}m$ |
|                                | LaMnO <sub>3</sub> | Mp-17554             | $Pnma$       |

(b)

**Protocol:**

1. Calculations are performed in **VASP** using the **PBE** functional. Projector Augmented Wave (PAW) potentials are applied with a consistent kinetic energy cutoff of 520 eV.
2. Enable D3 dispersion with Becke-Johnson damping (IVDW = 11).
3. Systematically scale the lattice parameters ( $a/a_0$ ) from **0.85 to 1.50** in fine steps of 0.02. This provides a Potential Energy Surface (PES) that crosses the D3 interaction cutoffs.
4. Extract the **Dispersion** Energy component ( $E_{\text{disp}}$ ) from each step. By analyzing the dispersion term alone, we remove the "electronic background" noise and isolate the D3 algorithm's behavior.
5. Construct a "Theoretical smooth  $1/R^6$  decay baseline by fitting a **interpolation** to the shoulders (smooth regions) of the curve and skipping the artifact region.
6. Error Measurement ( $\Delta E$ ): The error magnitude is the absolute vertical distance between the minimum of the artificial dip ( $E_{\text{min}}$ ) and the corresponding baseline value at that position ( $x_{\text{dip}}$ ).

$$\Delta E = E_{\text{min}} - E_{\text{baseline}}(x_{\text{dip}})$$

## Schematic:

(c)

| Material           | Space        | Dip Position | $E_{\min}$ (eV) | $E_{\text{expected}}$ (eV) | $\Delta E$ (eV) |
|--------------------|--------------|--------------|-----------------|----------------------------|-----------------|
| NiO                | $Fm\bar{3}m$ | 3.204        | -0.0311         | -0.0129                    | <b>-0.0182</b>  |
| CoO                | $Fm\bar{3}m$ | 3.469        | -0.0428         | -0.0191                    | <b>-0.0237</b>  |
| FeO                | $Fm\bar{3}m$ | 2.721        | -0.0910         | -0.0467                    | <b>-0.0443</b>  |
| TiO <sub>2</sub>   | $P4_2/mnm$   | 5.144        | -0.0332         | -0.0134                    | <b>-0.0198</b>  |
| RuO <sub>2</sub>   | $P4_2/mnm$   | 3.422        | -0.0470         | -0.0384                    | <b>-0.0087</b>  |
| IrO <sub>2</sub>   | $P4_2/mnm$   | 4.289        | -0.0339         | -0.0304                    | <b>-0.0035</b>  |
| SrTiO <sub>3</sub> | $Pm\bar{3}m$ | 7.174        | -0.1005         | -0.0388                    | <b>-0.0618</b>  |
| LaMnO <sub>3</sub> | $Pnma$       | 8.514        | -0.2222         | -0.0895                    | <b>-0.1326</b>  |

(d)

### 1. Variation Across Structure Types:

The results demonstrate a clear hierarchy: **Perovskites > Monoxides**

**> Rutilcs**. The highest artifact magnitude is seen in LaMnO<sub>3</sub> (0.133 eV).

This is a **artifact** of the D3 switching function. In the perovskite lattice, the A-site cation (La) is coordinated with 12 oxygens. During uniform expansion, all 12 bonds cross the  $D3$  cutoff threshold simultaneously.

### 2. Density and Regularity (Rutile Case):

The Rutile-type oxides (IrO<sub>2</sub>, RuO<sub>2</sub>) exhibit the lowest errors (under 0.01 eV). These are dense, edge-sharing octahedral networks. In such compact geometries, the local environments change more incrementally during expansion, allowing the  $D3, C_6$  coefficients to vary smoothly without the cascading dip seen in more open structures.

### 3. Influence of Ionic Size and Distortion:

Larger A-site cations (La<sup>3+</sup>, Sr<sup>2+</sup>) lead to larger unit cell volumes and more open coordination environments. Furthermore, the  $Pnma$  distortion in LaMnO<sub>3</sub> creates a variety of non-equivalent cation-oxygen bond lengths.

These staggered triggers make the  $D3$  coordination function particularly unstable, leading to the high measured error.

#### 4. The trend confirms that atomic weight is secondary to geometry.

$\text{IrO}_2$  contains the heaviest transition metal in the set but has the smallest artifact (0.0035) due to its rigid rutile packing. Thus,  $D3$ -induced errors are primarily **geometric glitches** that appear in low-density, high-coordination systems where the coordination-switching logic is stressed.

While geometry is the dominant factor, electronic properties dictate the severity of the artifact within a structural class. Highly polarizable ions, typical of early transition metals (like Ti in  $\text{TiO}_2$  and  $\text{SrTiO}_3$ ), generate large, unphysical negative energy drops when their local electron density is perturbed by strain. Conversely, while heavy ions like  $\text{Ir}^{4+}$  and  $\text{Ru}^{4+}$  are also highly polarizable, their compact rutile geometry "protects" them from exhibiting large  $D3$  artifacts. In the rock salt series, which generally shows near-zero variance, high-spin  $\text{Fe}^{2+}$  in  $\text{FeO}$  shows a slightly higher susceptibility to error than  $\text{NiO}$  because the Fe-O bond is more polarizable, making its local environment marginally more sensitive to strain-induced coordination changes before the cutoff is reached.